3.1. Experimental Design and Procedures. The initial design included seven treatments, broken into three broad categories by type of communication (no communication, structured communication, or chat). The six treatments with communication cross two types of communication (structured or free-form chat) with three different interaction structures between manager and agents (delegation with communication between agents, delegation with advice from managers, and managerial control). Before getting into the details of the experimental design, we pause to discuss the rationale for including both of these dimensions.

Structured communication and chat can be seen as extreme types of communication that serve different roles in our design.[1] Structured communication allows for clean tests of theoretical predictions since the message space matches the theoretical model being tested. More generally, the simplicity of structured communication makes it easier to pin down the mechanism by which communication affects outcomes. Free-form chat is richer and offers a more realistic form of communication. There exists ample evidence that free-form chat has differing effects from structured communication (Brandts et al., 2019), generally being more effective at promoting cooperation. There is no existing work that we know of comparing the effects of structured and free-form communication on either coordination or truth-telling.

Turning to the other dimension of the design, the first interaction structure, delegation with communication between agents, gives the manager no role. The latter two treatments, delegation with advice from managers and managerial control, explore different ways in which a manager might try to achieve efficient coordination, either persuading the agents to coordinate efficiently, essentially acting as a focal point, or imposing actions upon the agents. In the absence of asymmetric information, it is trivial to achieve efficient coordination by imposing actions. But it is no longer obvious which approach will be most effective with the addition of asymmetric information. Our treatments emphasize different approaches to the manager's problem, letting us see which will be more effective.[2]

The seven initial treatments are as follows:

No Communication – Delegation (NC – D): This was the baseline treatment where subjects played the MA game with delegation, as described in Section 2.2, without any additional communication.

Structured Communication between Agents – Delegation (SC/S – D): This treatment was identical to the **NC – D** treatment, except pre-play communication between agents was added. Prior to the agents' choices of actions, each game began with three rounds of messages. Within each round of messages, the agents simultaneously chose a pair of messages suggesting actions for themselves and the other agent. The

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[1] Section 5 presents a new treatment where the mode of communication is intermediate between these two extremes.

[2] The treatments with structured communication separately identify the effects of pre-play communication between agents and managerial advice. Given that neither had much effect in isolation, it seems safe to assume that the combination, paralleling **CH/A – D**, would also have little effect.

message space was limited in structured communication treatments; in **SC/S – D**, for example, the agents chose messages by clicking on radio buttons labeled with the five available actions and could not send any other messages. Agents observed each other's messages at the end of each round of messages. The purpose of having three rounds of messaging (rather than one) was to make it easier for agents to agree upon a course of action.

Structured Communication with Advice – Delegation (SC/A – D): This treatment was identical to the **NC – D** treatment, except the manager sent a message to the agents prior to each round of play. This message suggested actions for both agents in each of the five possible games. In other words, the manager advised a course of action contingent on the realized state of the world. The full message (a 5 x 2 matrix) was shown to both S1 and S2 prior to their choice of actions. The agents knew that both received identical messages.

Structured Communication – Managerial Control (SC – MC): In this treatment, subjects played the game with managerial control as described in Section 2.3. In each round, the two agents viewed the state of the world (i.e., Game 1, Game 2, etc.) and sent simultaneous messages to the manager reporting the state of the world ($\mu_i \in \{1,2,3,4,5\}$). There was no requirement that these messages be truthful, a point emphasized in the instructions. After receipt of the two messages, M chose both a row and a column. There was no requirement that the row and column match.

Chat between Agents – Delegation (CH/S – D): This treatment was identical to the **NC – D** treatment, except pre-play chat between agents was added. The agents had two minutes to engage in free-form discussions via chat before choosing actions. They could discuss whatever they chose. In practice, discussions largely focused on the obvious topic, how to play the game. The manager saw the discussion but could not participate.

Chat with Advice – Delegation (CH/A – D): This treatment was identical to the **CH/S – D** treatment, except the manager could participate in the chat prior to her agents choosing actions. As in **SC/A – D**, the manager had no control over the agents' actions but could advise them. Free-form chat opened many possibilities beyond simply advising a course of action, such as explaining the rationale for adopting efficient coordination or persuading agents of the mutual benefits of trusting each other.

Chat – Managerial Control (CH – MC): This treatment was identical to the **SC – MC** treatment, except the structured messages about the state of the world were eliminated and replaced by free-form chat between the agents and their manager. The agents were not specifically instructed to share information about the state of the world, but this was a natural and typical topic of conversation, making the structured messages redundant. Again, there was a two-minute time limit. Unlike **CH/A – D**, the manager had control over the outcome and the agents had reason to not truthfully reveal the state of the world.

Beyond the seven treatments reported in the main text, we ran an additional four treatments. These were modifications of the **NC – D** and **SC – MC** treatments to examine secondary issues. The main experimental design holds incentives fixed to focus on the effects of changing decision-making rights and the types of available communication. The **HSL – D** and **HSL – MC** treatments examine the effect of changing incentives by increasing state losses ($k_4 = 6$ vs. $k_4 = 4$). This reduces the tension between agents, making the efficient outcome more attractive relative to the safe outcome. Play shifts towards efficient coordination, consistent with the change in incentives, but our qualitative conclusions are unaffected. In particular, lowering state losses does not significantly increase efficiency gains relative to **SC – MC** and efficiency gains remain significantly lower than in **CH – MC**. In other words, a strong increase in incentives to play the efficient equilibrium has significantly less impact than allowing free-form communication. The **STR – D** and **STR – MC** treatments used strangers matching rather than partners matching. This made efficient coordination harder, as expected, but the effects are not significant and our qualitative conclusions are unaffected. Appendix D provides more description of these four additional treatments and the results.

We used a between-subjects design, so each subject participated in just one of the treatments. There were three sessions per treatment and nine three-person groups per session, giving 27 subjects per session, 27 independent groups per treatment, and a total of 567 subjects in 189 independent groups.

Subjects played 18 rounds in all treatments. They were assigned the role of M, S1, or S2 at the beginning of the session and kept these roles throughout the session. Partners matching was used (i.e. participants were matched with the same two subjects throughout the entire experiment). In treatments with delegation, the participants in the M role were pure observers. We did this to keep the possible influence of other-regarding preferences constant across treatments.

The state of the world (i.e. the game being played) was randomly and independently determined at the beginning of each round. Common seeds were used across treatments, limiting the possibility that treatment effects could be driven by differing draws. At the end of each round, subjects received feedback about the realized game and the chosen actions. In the treatments with managerial control, this made it possible for managers to know if an agent had lied about the game being played.

Sessions were run using z-tree (Fischbacher, 2007). Each session began with instructions (see Appendix C). Participants had printed copies of the payoff tables for all five games. Sessions were run at the LINEEX lab at the University of Valencia, with undergraduate students from the university as participants. The payoffs were denominated in Experimental Currency Units, with 1 ECU = 0.2€. Participants received their cumulative earnings for all rounds. Including a 5€ show-up fee, average pay was 19.90€. Sessions lasted approximately an hour.